OS Lab

Lab – 4

Shell Script (Part – 2)

Topic: Conditional Statements, User Input & Operations

Date: 02/02/26

Comparison Operators (if–else)

Symbol	Meaning	Operator (Shell Script)
==	Equal	-eq
≥	Greater than or equal	-ge
>	Greater than	-gt
<	Less than	-lt
≤	Less than or equal	-le
≠	Not equal	-ne

---

1. Create Script File

touch input.sh
nano input.sh

---

2. Conditional Statement (Fixed Value)

```
count=15
if [ $count -gt 12 ]
then
   echo "Condition is true"
else
   echo "Condition is false"
```

```
fi
```

---

3. Conditional Statement (User Input)

```
echo "Enter a number"
read count

if [ $count -gt 12 ]
then
   echo "Condition is true"
else
   echo "Condition is false"
fi
```

---

4. Create File Using Variable

```
echo "Enter your file name"
read name
echo "I will create a file called ${name}"
touch "${name}.sh"
```

---

5. Direct Symbol Use

```
echo "Enter a number"
read sum

if (( $sum > 20 ))
then
   echo "Condition is true"
else
   echo "Condition is false"
fi
```

---

6. if – elif – else Condition

```
echo "Enter a number"
read num

if [ $num -eq 0 ]
then
   echo "Number is zero"
elif [ $num -gt 0 ]
then
   echo "Number is Positive"
else
   echo "Number is Negative"
```

```
fi
```

---

7. Word Check (Static)

```
word=snow

if [ $word == "now" ]
then
   echo "true"
else
   echo "false"
fi
```

---

8. Word Check (User Input)

```
echo "Enter something"
read word

if [ $word == "x" ]
then
   echo "true"
elif [ $word == "y" ]
then
   echo "true"
else
   echo "false"
fi
```

---

9. Division of Two Numbers

```
echo "Enter two numbers"
read a b

if [ $b -eq 0 ]
then
   echo "Error: Division by zero is not possible"
   exit 1
fi

result=$(echo "scale=3; $a/$b" | bc)
echo "Division is: $result"
```

হলোBasic Calculator .

TASK – 1

Vote Eligibility Check

Condition:
Age ≥ 18 and Age < 70

```
#!/bin/bash

echo "Enter your age"
read age

if [ $age -ge 18 ] && [ $age -lt 70 ]
then
   echo "ELIGIBLE FOR VOTE"
else
   echo "NOT ELIGIBLE FOR VOTE"
fi
```

---

2. if–else Statement

C Language

```
#include <stdio.h>

int main() {
    int count = 15;

    if (count > 12) {
        printf("Condition is true");
    } else {
        printf("Condition is false");
    }

    return 0;
}
```

---

3. User Input Comparison

C Language

```
#include <stdio.h>

int main() {
    int num;
    printf("Enter a number: ");
    scanf("%d", &num);

    if (num > 10)
        printf("Greater than 10");
    else
        printf("Less than or equal to 10");

    return 0;
}
```

---

4. if – else if – else

C Language

```
if (num == 0) {
    printf("Zero");
}
else if (num > 0) {
    printf("Positive");
}
else {
    printf("Negative");
```

```
}
```

---

5. Logical AND Condition

C Language

```
if (age >= 18 && age < 70) {
    printf("Eligible for vote");
} else {
    printf("Not eligible for vote");
}
```

---

6. Word / String Comparison

C Language

```
#include <stdio.h>
#include <string.h>

int main() {
    char word[] = "x";

    if (strcmp(word, "x") == 0)
        printf("True");
    else
        printf("False");

    return 0;
}
```

---

9. Vote Eligibility Check (Shell vs C)

Problem Statement

Age **18 এর বেশি বা সমান** এবং **70 এর কম** হলে vote দিতে পারবে।

C Language

```c
#include <stdio.h>

int main() {
    int age;

    printf("Enter your age: ");
    scanf("%d", &age);

    if (age >= 18 && age < 70) {
        printf("ELIGIBLE FOR VOTE");
    } else {
        printf("NOT ELIGIBLE FOR VOTE");
    }

    return 0;
}
```

Difference Between Shell Script and C Language

Sl No	Topic	Shell Script	C Language
1	**Purpose**	OS automation, file handling, system tasks	Application & system programming
2	**Execution**	Interpreted (line by line)	Compiled (using compiler like gcc)
3	**File Extension**	.sh	.c
4	**Starting Line**	#!/bin/bash	#include <stdio.h>
5	**Variable Declaration**	No data type needed (e.g., x=10)	Data type required (e.g., int x = 10;)
6	**User Input**	read var	scanf()
7	**Output**	echo	printf()
8	**Comparison Operators**	-gt, -lt, -eq, etc.	>, <, ==, etc.
9	**Logical Operators**	&& (AND), ` OR, NOT	AND, OR, NOT
10	**Conditional Statement**	if [condition] then ... fi	if(condition) { ... }
11	**else if**	elif	else if
12	**Loop Support**	Limited (for, while)	Powerful (for, while, do-while)
13	**String Comparison**	== inside [ ]	strcmp() function
14	**Arithmetic Operation**	expr, bc, $(( )), Direct operators +, -, *, /	Direct operators +, -, *, /
15	**Speed**	Slower (Interpreted)	Faster (Compiled)
16	**Error Handling**	Simple	Advanced
17	**Use Case**	Automation, server scripting	Software & OS development